

Problem 1: Use five counters in a row. Write R as 0 and B as 1. A score counts matching positions. Build each secret below and score all four tests. Keep the secret visible while checking.

Secret	00000	00011	00101	01001
00000				
11111				
10010				

Problem 2: Choose a new secret and hide it. Your partner uses exactly the same four tests. Record the scores. Reveal the secret and check each score. Then swap jobs.

Round 1		Round 2	
Test	Score	Test	Score
00000		00000	
00011		00011	
00101		00101	
01001		01001	

Revealed secrets:

Problem 3: Change two test positions from 00 to 11 while leaving all other positions fixed. Build each possible secret pair below. Count its matches with 00 and with 11, and record the change.

Secret pair	Matches with 00	Matches with 11	Score change
00			
01			
10			
11			

Problem 4: For one hidden five-place secret, tests 00000, 00011, 00101, and 01001 scored 2, 2, 2, and 0. Use the first score to count its 1s. Compare each later score with the first to find the number of 1s in each changed pair.

Positions	Number of 1s
All five positions	
Positions 4 and 5	
Positions 3 and 5	
Positions 2 and 5	

Problem 5: Use those counts to build the secret. Record it and check all four scores directly.

My secret and its four scores:

Problem 6: Try to build a secret with exactly one 1 in each of these pairs: positions 4 and 5, positions 3 and 5, positions 2 and 5. First put 0 in position 5, then put 1 there. In each case try both choices for position 1. Record all four words and their total numbers of 1s.

Position 5	Position 1	Five-place word	Total 1s
0	0		
0	1		
1	0		
1	1		

Problem 7: Score your four words using the four tests from Problem 1. Compare the score records. Circle a score that separates each pair of words.

Secret	00000	00011	00101	01001

Problem 8: Return to your two hidden secrets from Problem 2. Use the total number of 1s and the three pair counts to recover them from the scores alone. Record the recovered words.

Recovered words: